Cardiovascular System (CARD) · NPLEX-P1-CARD-04

CV-4 Heart Valves and the Cardiac Cycle

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-04-01-C01				
NPLEX-P1-CARD-04-01-C02				
NPLEX-P1-CARD-04-01-C03				
NPLEX-P1-CARD-04-01-C04				
NPLEX-P1-CARD-04-01-C05				
NPLEX-P1-CARD-04-01-C06				
NPLEX-P1-CARD-04-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C01-BD1

Draw the four valves and where each is heard

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C01 Activity: NPLEX-P1-CARD-04-01-C01-BD1 Date: _____

Without looking at your notes, draw the four heart valves in position and build a table beside the drawing with one row per valve. For each valve write the chambers or vessels it separates, the number and names of its cusps, what holds it closed, and the chest location where it is best heard with the intercostal space. Then write two sentences on what the fibrous skeleton does for the valves and two sentences on why a valve cusp moves only when the pressure across it changes. Finish by writing where the murmur of aortic stenosis radiates and where the murmur of mitral regurgitation radiates, and give the anatomical reason for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C01-BD2

Why an inferior infarct tears one papillary muscle and not the other

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C01 Activity: NPLEX-P1-CARD-04-01-C01-BD2 Date: _____

A man has an occlusion of his posterior descending artery and five days later develops sudden breathlessness with a new loud murmur at the apex. Write which papillary muscle failed and why that one and not the other, naming the blood supply of each left ventricular papillary muscle. Then write in order what happens from the moment the muscle gives way: what the mitral cusps do during systole, where the blood goes, what the left atrial pressure does, and what appears in the lungs and in the forward stroke volume. Write why this is tolerated badly compared with mitral regurgitation that has built up over years. Finish by writing two other ways the same loss of cusp support occurs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-04-01-C01-CA01

Sudden breathlessness five days after an inferior infarction

A 66-year-old man was admitted five days ago with 4 hours of chest pain and ST elevation in leads II, III, and aVF. He improved, and this evening he becomes acutely breathless and cannot lie flat. Temperature is 37.1 degrees C (98.8 degrees F), pulse is 112/min and regular, respirations are 30/min, and blood pressure is 86/60 mm Hg. A new holosystolic murmur is heard at the apex and radiates to the left axilla. Crackles are present through both lung fields. The jugular venous pulsation is raised. A pressure tracing from the pulmonary artery wedge position shows a small a wave and a very large wave in late systole, with a mean wedge pressure of 30 mm Hg. The electrocardiogram is unchanged from this morning.

1. Recognize. Name the structural complication, the valve involved, and the artery whose occlusion led to it.

2. Explain. Explain the mechanism from the blood supply of the affected structure through to the murmur, the wedge tracing, and the pulmonary crackles.

3. Differentiate. Write how this presentation differs from a ventricular septal rupture and from free wall rupture with tamponade, naming the findings on examination and on the wedge tracing that separate them.

4. Investigate. Write which measurements on the wedge and jugular tracings you would look at to support your answer, and what each one would show here.

5. Decide. State why the blood pressure is low even though the total volume the ventricle ejects each beat has risen.

6. Justify. Justify that answer from where the regurgitant volume goes, from the compliance of an atrium that has had no time to dilate, and from what has happened to the forward stroke volume.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C02-BD1

Draw the Wiggers diagram and mark every valve event

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-BD1 Date: _____

On one set of axes with time across the bottom, draw left atrial pressure, left ventricular pressure, and aortic pressure, and underneath them draw left ventricular volume, the ECG, and the heart sounds. Mark the four valve events on the tracing and write beside each one which two pressures have just crossed and in which direction. Label isovolumetric contraction, rapid ejection, reduced ejection, isovolumetric relaxation, rapid filling, diastasis, and atrial systole. Write the normal numbers on the drawing: aortic diastolic and peak pressures, end-diastolic and end-systolic volumes, stroke volume, ejection fraction, pulmonary artery pressure, and the length of one cycle with its systolic and diastolic parts at 75 beats per minute. Finish by marking where S1 and S2 fall and writing what is vibrating to make each one.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C02-BD2

Name the phase from four numbers, and what a fast rate costs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-BD2 Date: _____

Three tracings are sampled at one instant each. Tracing A: left atrial pressure 9 mm Hg, left ventricular pressure 45 mm Hg, aortic pressure 84 mm Hg, left ventricular volume 120 mL. Tracing B: left atrial pressure 9 mm Hg, left ventricular pressure 45 mm Hg, aortic pressure 84 mm Hg, left ventricular volume 50 mL. Tracing C: left atrial pressure 12 mm Hg, left ventricular pressure 6 mm Hg, aortic pressure 78 mm Hg, left ventricular volume 70 mL. For each tracing write the state of the mitral valve and the aortic valve, name the phase, and show the reasoning as a comparison of two pressures at a time. Write which single measurement separates tracing A from tracing B and why pressure cannot do it. Then the heart rate rises from 75 to 160 beats per minute. Write what happens to the length of systole, the length of diastole, the part of diastole that is lost first, and ventricular filling, and explain why left ventricular coronary flow suffers. Finish by writing why the pulmonary valve opens before the aortic valve and closes after it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-04-01-C02-RT01

Rebuild the valve events of one left-sided cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-RT01 Date: _____

Rebuild the nine-step sequence of valve events in one left-sided cardiac cycle from atrial systole through to diastasis, one step per line and in order. Beside each step that is a valve opening or closing, write the two pressures being compared and the approximate pressure at which it happens. Mark the two steps during which every valve is closed and write what the ventricular volume is doing in each. Mark the step that produces S1 and the step that produces S2, and beside the aortic closure step write what makes the dicrotic notch on the aortic curve. Finish by writing what makes S1 loud and what makes it soft, with the reason in each case.

Rebuild from memory: Valve events in one left-sided cardiac cycle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C03-BD1

Draw the normal pressure-volume loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C03 Activity: NPLEX-P1-CARD-04-01-C03-BD1 Date: _____

Draw the left ventricular pressure-volume loop with pressure on the vertical axis and volume on the horizontal axis. Mark the direction the loop is traced. Label all four corners with the valve event that happens there and write the pressure and volume at each one. Label the two vertical sides and the top and bottom edges with the phase each represents. Then mark on the drawing what the width of the loop measures and what the enclosed area measures, and write beside the bottom edge and the top left corner which relationship each one follows and what changes its slope. Finish by writing the normal end-diastolic volume, end-systolic volume, stroke volume, and ejection fraction range.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C03-BD2

Name the lesion from the loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C03 Activity: NPLEX-P1-CARD-04-01-C03-BD2 Date: _____

Three loops are described. Loop 1: end-diastolic volume 190 mL, end-systolic volume 65 mL, peak pressure 115 mm Hg, and the left side of the loop slopes to the right so volume rises while pressure is falling after ejection. Loop 2: end-diastolic volume 125 mL, end-systolic volume 75 mL, peak pressure 215 mm Hg with aortic pressure 115/70 mm Hg, and a steep bottom edge with an end-diastolic pressure of 22 mm Hg. Loop 3: end-diastolic volume 200 mL, end-systolic volume 70 mL, peak pressure 105 mm Hg, and volume starts falling at the bottom right corner while pressure is only about 15 mm Hg. For each loop calculate the stroke volume, state whether the load is a pressure load or a volume load, name the lesion, and write the one feature of the loop that settles it. For each lesion write which isovolumetric phase is no longer isovolumetric and why, which pattern of hypertrophy follows, and what has happened to the forward stroke volume compared with the total. Finish by writing what the loop of significant mitral stenosis looks like and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-04-01-C03-CP01

Pressure-volume loop in three valve lesions

Fill in every cell from memory, then open the notes and correct any cell you missed.

Feature	Aortic stenosis	Aortic regurgitation	Mitral regurgitation
Pressure load or volume load			
Peak left ventricular pressure, and how it compares with aortic pressure			
End-diastolic volume			
End-systolic volume			
Total stroke volume, and what happens to the forward volume			
Which isovolumetric phase is no longer isovolumetric, and what the volume does during it			
Pattern of chronic hypertrophy			
Shape and position of the loop			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-04-01-C03-Q01

A loop that loses volume before the aortic valve opens

A pressure-volume loop recorded from the left ventricle of a 58-year-old woman shows an end-diastolic volume of 195 mL, an end-systolic volume of 70 mL, and a peak ventricular pressure of 108 mm Hg. Volume begins to fall at the bottom right corner of the loop while ventricular pressure is still only 16 mm Hg, so the right side of the loop slopes to the left rather than running vertically. Her measured forward cardiac output is below normal. Which of the following best explains the shape of this loop?

A. A fixed obstruction to left ventricular outflow raising the pressure needed to eject

B. Reduced left ventricular compliance with a steep end-diastolic pressure-volume relationship

C. Blood escaping into the left atrium from the onset of ventricular contraction

D. Blood returning from the aorta into the ventricle during relaxation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C04-BD1

Draw the jugular venous pulse against the heart sounds

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C04 Activity: NPLEX-P1-CARD-04-01-C04-BD1 Date: _____

Draw one cycle of the jugular venous pulse and underneath it draw the ECG and mark S1 and S2. Label the a wave, the c wave, the x descent, the v wave, and the y descent. Beside each label write the mechanical event in the right atrium or ventricle that produces it and its timing relative to the P wave, S1, or S2. Then write how the jugular pulse is told apart from the carotid pulse, listing four differences. Finish by writing how the height of the pulsation is used to estimate right atrial pressure, including the distance from the sternal angle to the right atrium, the angle at which the patient sits, and the height above the sternal angle that suggests raised pressure.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C04-BD2

Read four abnormal neck tracings

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C04 Activity: NPLEX-P1-CARD-04-01-C04-BD2 Date: _____

Four patients have abnormal jugular findings. Patient 1 has a pulse of 36 beats per minute and occasional very large pulsations that do not appear with every beat. Patient 2 has a narrow complex tachycardia with large pulsations that appear with every beat. Patient 3 has an irregularly irregular pulse and no a wave at all. Patient 4 has a large systolic wave with no x descent, a steep y descent, and a pulsating liver. For each patient name the abnormal wave or missing wave, write the mechanism in terms of what the right atrium is contracting or filling against, and write the rhythm or lesion responsible. For patient 1 explain why the large pulsations come only on some beats and for patient 2 explain why they come on every beat. For patient 4 write what has merged with what and why the pressure rises right through systole, then write which tracing shows the same picture in acute mitral regurgitation. Finish by writing what a large a wave that is not a cannon wave tells you and give two causes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-04-01-C04-Q02

Intermittent giant neck pulsations

A 77-year-old woman comes to the emergency department after two episodes of collapse. Pulse is 40/min and regular, and blood pressure is 118/72 mm Hg. Inspection of the neck shows occasional very large venous pulsations that are not palpable and do not occur with every beat. The electrocardiogram shows P waves at a rate of 92/min and QRS complexes at a rate of 40/min with no fixed relationship between them. Which of the following best explains the large venous pulsations?

A. Right atrial contraction against a closed tricuspid valve on beats where the timing coincides

B. Regurgitation of blood from the right ventricle into the right atrium through an incompetent tricuspid valve

C. Obstruction of right atrial emptying through a stenotic tricuspid valve

D. Transmitted pulsation from the adjacent carotid artery during ventricular ejection

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C05-BD1

Build the S2 splitting table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-BD1 Date: _____

On blank paper build a table of the four patterns of S2 splitting with one row each for physiologic, wide, fixed, and paradoxical. Use the columns order of the two components, what is heard in expiration, what is heard in inspiration, the mechanism, and the classic causes, and fill every cell. Above the table write which component of S2 is loud and heard widely, which is soft, where splitting is listened for, and how far apart the components must be for a split to be audible. Beside the physiologic row, draw the timing of the two components in expiration and in inspiration. Finish by writing the two questions you ask at the bedside to place any S2 into one of these four rows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C05-BD2

Four patients with a split S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-BD2 Date: _____

Four patients have an abnormal second sound. Patient 1 has a split heard in expiration that widens further on inspiration, and the ECG shows an rSR prime in V1 with a QRS of 0.14 s. Patient 2 has a wide split that does not change at all through the respiratory cycle. Patient 3 has a split heard in expiration that becomes single on inspiration, and an ejection murmur at the right second intercostal space with a soft A2. Patient 4 has a split heard in expiration that widens on inspiration, with a holosystolic murmur at the apex radiating to the axilla. For each patient name the pattern, write which component is early or late and why, and write whether the respiratory mechanism is still working. For patient 2 explain from the behaviour of the two atria why breathing no longer changes the split. For patient 3 explain why inspiration narrows rather than widens it. For patient 4 explain which component has moved and what gave the left ventricle a second route out. Finish by writing the two mechanisms, one electrical and one mechanical, that delay a component, and give an example of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-04-01-C05-RT01

Rebuild the sequence that widens the split on inspiration

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-RT01 Date: _____

Rebuild the seven-step sequence that runs from the start of inspiration to a wider split of S2, one step per line and in order. Mark which steps act on the right heart and which act on the left. Beside the step about pulmonary vessels, write what happens to pulmonary vascular impedance and what that does to the length of right ventricular ejection. Beside the step about left ventricular filling, write why filling dips and what that does to the timing of A2. Underneath, draw A2 and P2 on a timeline for expiration and for inspiration with the interval written in seconds, and write in which age group this is easiest to hear.

Rebuild from memory: Physiologic splitting of S2 during inspiration

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C06-BD1

S3 and S4 side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C06 Activity: NPLEX-P1-CARD-04-01-C06-BD1 Date: _____

Write out S3 and S4 side by side on one page. For each sound give its timing in diastole, the interval from S2 where the notes gave one, the mechanical event that produces it, what is vibrating, where and with which part of the stethoscope it is heard on the left and on the right, and what makes a right-sided one louder. Then for each sound list when it is a normal finding and when it means disease, with the conditions from the notes. Write what happens to each sound in atrial fibrillation and give the reason. Finish by writing why significant mitral stenosis prevents an S3.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C06-BD2

Decide which murmur is just flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C06 Activity: NPLEX-P1-CARD-04-01-C06-BD2 Date: _____

Write the physical reason a normal valve can produce a murmur, naming the quantity that predicts turbulence and the three things that raise or lower it. Then take four people: a woman at 30 weeks of pregnancy, a man with a haemoglobin of 68 g/L, a child of six with a thin chest, and a woman with a temperature of 39.4 degrees C (103 degrees F). For each one write which part of that quantity has changed and in which direction, and what happens to the murmur when the cause is removed. Next list the features that let you call a murmur functional: grade, timing, location, what S2 does, and what extra sounds are present. Then list the findings that argue for structural disease instead, and name the one continuous murmur that is still innocent along with the two bedside moves that abolish it. Finish by writing the three murmurs a child commonly has and where each is heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C07-BD1

Build the murmur timing table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C07 Activity: NPLEX-P1-CARD-04-01-C07-BD1 Date: _____

Build a table of murmurs from memory with one row per lesion, using the columns timing and shape, lesion, where it is best heard, and the one feature that gives it away. Cover aortic stenosis, pulmonic stenosis, hypertrophic cardiomyopathy, mitral regurgitation, tricuspid regurgitation, ventricular septal defect, mitral valve prolapse, aortic regurgitation, mitral stenosis, tricuspid stenosis, and patent ductus arteriosus. Above the table write the rule that places any murmur in systole or diastole from the valve involved, and write the grading scale from 1 to 6 with the grade at which a thrill becomes palpable. Underneath, write why an ejection murmur starts after S1 rather than with it, why a holosystolic murmur runs the whole of systole, and why the murmur of mitral valve prolapse comes late with a click in front of it. Finish by writing what happens to the peak of the aortic stenosis murmur and to A2 as the stenosis worsens, and what the interval from S2 to the opening snap says about the severity of mitral stenosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C07-BD2

Separate two systolic murmurs with the bedside maneuvers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C07 Activity: NPLEX-P1-CARD-04-01-C07-BD2 Date: _____

Write what each of these changes at the bedside: inspiration, squatting, standing from a squat, handgrip, and Valsalva strain. For each one write whether preload, afterload, or both are altered and in which direction. Then write what most murmurs do with more preload and what regurgitant murmurs do with more afterload, and the reason in each case. Two patients each have a systolic murmur at the left sternal border that grows louder when they stand. Write which two conditions behave this way, what is happening inside the left ventricle as the cavity gets smaller, and what each finding does on squatting and on handgrip. Finish with a third patient who has an ejection murmur radiating to the carotids that gets softer on standing and softer on handgrip, and write why a fixed obstruction behaves the opposite way to the first two.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-04-01-C07-PR01

Predict what standing does to a click and a late systolic murmur

A 29-year-old woman has a midsystolic click followed by a late systolic murmur at the apex. She stands up quickly from a squat. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return and to left ventricular end-diastolic volume		
What happens to the size of the left ventricular cavity at the start of systole		
What happens to the point in systole at which the mitral cusps give way		
What happens to the timing of the click		
What happens to the length and loudness of the murmur		
What happens to the click and the murmur when she squats again and when she performs handgrip		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____